

Harry Addison

+44 7765671466 | addisonh@sky.com | [harryaddison.github.io](https://github.com/harryaddison)

 [Harry Addison](#) |  [HarryAddison](#) |  0000-0001-8271-1364

Dagenham, Essex, RM10 8QA, United Kingdom

OBJECTIVE

Seeking a position in transient astronomy/astrophysics to leverage my expertise in large scale surveys. Aiming to contribute to innovative projects that further our understanding of supernova explosion mechanisms, progenitor systems, and improving their reliability as probes of binary evolution, galaxy evolution, and cosmology.

EXPERIENCE

- **University of Surrey** September 2022 - Current
PhD Researcher Guildford, UK
 - Developed pipeline to analyse ~25,000 supernova spectra from the Time-Domain Extragalactic Survey (TiDES)
 - Investigated selection of early-time supernova from all-sky surveys for spectroscopic follow-up with TiDES
 - Developed analysis software for the commissioning of the 4m Multi-object spectroscopic telescope (4MOST)
 - Conducted analysis of 4MOST commissioning data, specifically monitoring the health of the instrument.

EDUCATION

- **University of Surrey** September 2022 - Current
PhD Guildford, UK
 - Thesis Title: Supernova with the Time-Domain Extragalactic Survey and the commissioning of 4MOST
 - Supervisors: Bob Nichol - University of Surrey; Denis Erkal - University of Surrey; Chris Frohmaier - University of Portsmouth
- **University of Surrey and Radboud University** September 2018 - July 2022
Masters in Physics with Astronomy Guildford, UK; Nijmegen, Netherlands
 - Grade: First Class Honours
 - Thesis Title: Identifying Galactic Luminous Red Nova Precursors
 - Supervisors: Nadejda Blagorodnova - University of Barcelona, Institute of Space Studies of Catalonia; Paul Groot - Radboud University, University of Cape Town, South African Astronomical Observatory

PROJECTS

- **4MOST on-site commissioning** November 2025
Tools: Python, SAOImageDS9, 4MOST reduction pipelines, VISTA and 4MOST control software
 - Involved in the commissioning of the 4m Multi-Object Spectroscopic Telescope (4MOST) on the VISTA telescope
 - Took responsibility for 3 procedures regarding the calibration frames (darks, bias, and flat frames), ensuring the procedures feasibility, developing analysis scripts, and overseeing the data reductions and analysis specific for these procedures, providing detailed reports of all findings
 - Operated 4MOST/VISTA: taking calibrations, science observations, and performing system tests and experiments to determine any limitations or errors with 4MOST/VISTA
 - Reduced and analysed data that I had observed, ensuring that 4MOST was performing as expected.
 - Solving issues that arose during observing and data reductions/analysis
- **Supernova Analysis Pipeline:** January 2024
Tools: Python, machine learning  [GitHub Repository](#)
 - Developed pipeline for TiDES to measure equivalent widths and velocities from ~30,000 supernova spectra
 - Investigated use of machine learning methods for providing measurements, concluding that we need larger datasets such as one that TiDES will produce
 - Applied Gaussian Process Regression and Principal Component Analysis to be able to take measurements of low signal to noise (SNR~1) spectra
 - Implemented the pipeline into the TiDES data flow and infrastructure using Docker, allowing for adaptive scalability
- **4MOST commissioning software development:** October 2023
Tools: Python, SAOImageDS9
 - Developed a software package for handling 4MOST commissioning data, specifically focusing on calibration frames (dark and bias frames)
 - Worked as a team of 6 for 3 weeks to supply a working prototype to our user-base.
 - Presented and demonstrated our working product to the user-base, gaining positive feedback

• Search for Galactic luminous red novae:

Tools: Python, machine learning

February 2021

 [GitHub Repository](#)

- Developed a novel method to identify Galactic Luminous Red Novae (LRNe) before their outburst
- Applied Gaussian Mixture models to Gaia observed colour magnitude diagrams to identify Hertzsprung-gap stars (yellow giants/supergiants), which are believed LRNe progenitors
- Data mined archival Zwicky Transient Facility time-series data, applying an algorithm to identify LRN-like variability in the previously identified Hertzsprung-gap stars
- Identification of 21 LRNe candidates, one of which was later determined to be a high mass X-ray binary system, which exhibit similar time-series features as pre-outburst LRNe

PUBLICATIONS

J=JOURNAL, S=IN SUBMISSION, O=OTHER

- [J] C. Frohmaier, M. Vincenzi, M. Sullivan, S.F. Hönig, M. Smith, H. Addison, et al. (2025). **TiDES: The 4MOST Time Domain Extragalactic Survey**. *The Astrophysical Journal*, Vol. 992, Issue 1, pp. 158. DOI: 10.3847/1538-4357/adff4e
- [O] A. Reguitti, P. Ochner, H. Addison, et al. (2025). **Padova-Asiago Transient Classification Report for 2025-05-14**. *Transient Name Server Classification Report*, No. 2025-1785.
- [S] H. Addison, C. Frohmaier, K. Maguire, et al. (2024). **TiDES – Young Supernova Selection Pipeline**. Manuscript submitted for publication in *MNRAS*. arXiv:2412.05364
- [J] H. Addison, N. Blagorodnova, P.J. Groot, et al. (2022). **Searching for the next Galactic Luminous Red Nova**. *MNRAS*, Vol. 517, Issue 2, pp. 1884-1900. DOI: 10.1093/mnras/stac2685

TALKS / CONFERENCES

T=TALK, P=POSTER

- [T] Harry Addison (2025). **Commissioning the 4MOST Telescope: Enabling Next-Generation Spectroscopic Surveys**. *University of Surrey Maths and Physics PGR Conference*, Contributed talk.
- [T] Harry Addison (2024). **Early-Time Supernova Spectroscopy with TiDES - 4MOST**. *UK Transients Meeting*, Contributed talk.
- [T] Harry Addison (2023). **Using Lasair to select early-time transients for follow-up**. *Lasair User Meeting*, Contributed talk.
- [T] Harry Addison, et al. (2023). **Search for Galactic Luminous Red Novae**. *EAS2023, European Astronomical Society Annual Meeting*, Session S11 : Stellar interactions: contact binary stars and common-envelope evolution, Contributed talk, id. 1786.
- [P] Harry Addison, et al. (2023). **Identifying Galactic Luminous Red Novae Precursors**. *EAS2022, European Astronomical Society Annual Meeting*, Session S12 : Gaia: The (TWO) Billion Star Galaxy Census: The Dawn of Gaia DR3, ePoster, id. 538

COMPUTING EXPERIENCE

- **Python (advanced):**
 - * Github: github.com/HarryAddison
 - * Gitlab (Private 4MOST codes): gitlab.4most.eu/addisonh
- **HTML, CSS (intermediate):**
 - * TiDES website Github repository: github.com/TiDES-4MOST/tides-website
 - * Personal website: harryaddison.github.io
 - * Personal website Github repository: github.com/HarryAddison/harryaddison.github.io

OBSERVING EXPERIENCE:

- **Proposals:**
 - * CoI of the “Constraining the progenitor systems of thermonuclear supernovae through early spectroscopy”, Program ID 117.29VT at the VLT telescope with XSHOOTER, submitted - waiting decision
- **Observing Nights:**
 - * 2025, The Galileo Telescope at the Astrophysical Observatory of Asiago, Boller and Chivens spectrograph, 3 nights at the telescope.
- **Data Reduction:**
 - * 4MOST Quality Control and Level 1 Data Reduction pipelines. Instrument: 4MOST. Expertise: Reduction of all data taken during commissioning of 4MOST, with specific expertise on day time calibrations (bias, dark and flat frame calibrations).

SCIENTIFIC COMMUNITY WORK:

- **4MOST's Science Communication working group**, February 2025 - present
 - * Website maintainer, social media content creator, news article writer, key member of 4MOST's first-light press release
- **Public Outreach: World Space Week**, October 2023, October 2024, October 2025
 - * Demonstrating my work to the public at Guildford's World Space Week events
- **Undergraduate teaching assistant**, University of Surrey, Dates: February 2023 - June 2023, September 2025 - January 2026
 - * Providing assistance to undergraduate students during computing lab sessions, answering questions regarding Python, physics, and specialised software (e.g SAOImageDS9).

PROFESSIONAL MEMBERSHIPS

- **4m Multi-Object Spectroscopic Telescope**, Permanent Science Team Member, 2022 - Permanent
- **Time-Domain Extragalactic Survey**, Permanent Membership, 2022 - Permanent

ADDITIONAL RESEARCH EXPERIENCE:

- **4MOST Observing School**, 04/2025
 - Astrophysical Observatory of Asiago, Italy.
 - Acquired training in spectroscopic observations and data reduction.
- **4MOST Commissioning Bootcamp**, 11/2024
 - Leibniz Institute for Astrophysics Potsdam (AIP), Germany.
 - Acquired training in 4MOST/VISTA observing procedures, 4MOST data reduction and analysis workflows, and collaborative team building with the commissioning team.
- **Astronomy Summer School**, 09/2024
 - University of Sussex, UK.
 - Acquired training in software used in extragalactic astronomy.
- **4MOST Commissioning Bootcamp**, 11/2023
 - Leibniz Institute for Astrophysics Potsdam (AIP), Germany.
 - Acquired training in 4MOST/VISTA observing procedures, 4MOST data reduction and analysis workflows, and collaborative team building with the commissioning team.
- **4MOST Commissioning Coding Campaign**, 10/2023 - 11/2023
 - Leibniz Institute for Astrophysics Potsdam (AIP), Germany.
 - Developed analysis software for use in the commissioning of 4MOST.
 - Acquired training in 4MOST data reductions and analysis, and higher level coding practices.

REFERENCES

- **Bob Nichol**
Pro-Vice-Chancellor and Executive Dean, Faculty of Engineering and Physical Sciences
University of Surrey
Email: bob.nichol@surrey.ac.uk
Phone: +44 1483683332
Relationship: PhD Supervisor
- **Kate Maguire**
Professor in Physics, School of Physics
Trinity College Dublin
Email: kate.maguire@tcd.ie
Relationship: International Collaborator
- **Chris Frohmaier**
LSST:UK Senior Research Fellow
University of Portsmouth
Email: chris.frohmaier@port.ac.uk
Relationship: PhD External Supervisor